MIL-P-46364A(MU)
13 November 1967
SUPERSEDING
MIL-P-46364(MU)
7 December 1962

MILITARY SPECIFICATION

PERISCOPE, BATTERY COMMAND: M43

1. SCOPE

1.1 This specification covers one type of optical mechanical instrument known as Periscope, Battery Command: M43.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

FSC 1240

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DRAWINGS

U. S. Army, Frankford Arsenal

F8565556	Pressure Tester, Two Station
F8623822	Periscope, Battery Command: M43
IEL 8623822	Inspection Equipment for Periscope, Battery Command: M43

PACKAGING DATA SHEET

8624251	Packaging of Periscope, B.C. M43

(Copies of specifications, standards, drawings, and packaging data sheets required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The periscope shall be manufactured in accordance with Drawing F8623822.

3.2 General specification.- The contractor shall be responsible for compliance with the following requirements of MIL-F-13926:

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking

3.3 Temperature conditions.- The requirements of this specification shall be met with the periscope thermally stabilized at an ambient tempera ture of between +60 and +90°Fahrenheit (F) unless otherwise specified in the requirement paragraphs.

3.4 Workmanship.- The workmanship requirements of MIL-F-13926 shall be met. The exterior protective surface finishes shall be free of any defects which effect their protective qualities or appearances. Optical surfaces shall be clean and free of any scratches or blemishes which may effect their optical qualities or appearance.

3.5 Environmental.- Any periscope subjected to any or all of the requirements of 3.5.1 to 3.5.4 inclusive shall subsequently meet the requirements of 3.6 to 3.7.35.1 inclusive, unless otherwise specified.

3.5.1 Storage temperature.- The periscope shall show no evidence of physical failure such as cement separation, glass breakage or casting failure after having been exposed and thermally stabilized at ambient temperatures of -65° and +165° Fahrenheit (F).

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3.5.2 Extreme operation temperatures.- The periscope shall meet the requirements of 3.7.31 while exposed and thermally stabilized at ambient temperatures of -40° and +150°F.

3.5.3 Vibration.- The periscope shall be vibrated in the Carrying Case M165. The case, exclusive of all accessory items, shall be positioned in its normal horizontal loading psotion, the cover closed, and the case secured to an adapter fixture designed to support the case on its concaved bottom surface. The assembly shall then be vibrated at a constant frequency of 30 cycles per second with a single amplitude of 0.031 inch in a vertical direction for a period of 2 to 2.5 minutes, at room temperature, 60° to 90°F.

3.5.4 Sealing and flushing.- The periscope shall be pressurized to an internal pressure of 5 psig, using dry nitrogen having a dew point at least as low as -25°F. The pressure shall be maintained at 5 psig for a period of five minutes to balance pressures within the instruments internal cells. Subsequent to the five minute balancing period the periscope shall show no evidence of leakage with an internal pressure of 5 psig over a period of 5 minutes ± 15 seconds. The internal pressure shall then be reduced to zero psig. The periscope shall then be flushed with dry nitrogen until the interior atmosphere of the periscope has a dew point at least as low as -25°F. The periscope shall then be sealed with an internal pressure of between 0.25 and 0.50 psig of dry nitrogen.

3.6 Cleanliness.- The periscope shall meet the cleanliness requirements of MIL-O-13830. There shall be no more than 2 particles of dirt visible on each of the 3 reticles and 2 scales. Each particle shall not be in excess of an area equivalent to a 0.004 inch diameter as indicated in the focal plane of the right eyepiece. There shall be no dirt on other optical surfaces of the periscope which interferes with observation scale or reticle readings, or adversely effects image quality.

3.7 Accuracy.

3.7.1 Orientation.- With the spindle axis of the telescope vertical within 0.1 mil, the azimuth (AZ) and elevation (EL) scales reading zero (0) and an incoming line of sight horizontal and filters in the clear position, the requirements of 3.7.1.1 to 3.7.29.3 inclusive shall be met.

3.7.1.1 Level vial readings.- The cylindrical level vials shall read central within the width of one vial graduation line when the line of sight through the boresight cross of the periscope reticle is in coincidence with a horizontal target line within a total deviation of ±0.2 mil, with the periscope rotated from 0 to 6400 mils in azimuth, by rotation of the periscope azimuth rotation knob.

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3.7.2 Collimation.- Parallel rays entering the objectives at zero elevation shall emerge from the eyepieces parallel within 15 minutes of arc in dipvergence, and shall not diverge less than 5 nor more than 30 minutes of arc. These conditions shall be met at infinity focus setting of the eyepieces, and for any setting of the interpupillary scale between 58 and 72 millimeters, at the high and low power positions.

3.7.3 Level line.- The line of sight through the geometric axis of the reticle shall be in coincidence with the incoming horizontal line of sight within 1.0 mil in elevation.

3.7.4 Field of view.- The centering error of the field of view (reticle pattern) in the right eyepiece relative to the right diaphragm shall not exceed 3.0 mils in azimuth and elevation as read on the reticle for the high and low power positions.

3.7.5 Resolution.- Target resolution of each telescope on the optical axis shall be 8 seconds of arc or less, for the high and lower power positions, with the filter knob at clear position, at one focus setting of the eyepiece.

3.7.6 Parallax.- Parallax between the target image and the center of the telescope reticle shall not exceed 0.1 mil, with a target at 1000 ± 100 meters.

3.7.7 Magnification.- Magnification of right and left optical system shall not differ from each other by more than one percent, for high and low power systems.

3.7.8 Filters.- There shall be complete filter coverage for each telescope except for filter position 5 which shall be clear. There shall be no overlap of adjacent filters in the field of view.

3.7.9 Reticle accuracy.- The angular separation between the left and right 40 mils graduation on the horizontal reticle line through the geometric axis of the reticle shall be 80 mils ± 0.5 mil.

3.7.10 Image tilt.- The images of a plumb line formed by the two optical systems shall be vertical within one degree and parallel within 0.5 degree.

3.7.11 Parallelism of reticle and image.- The vertical reticle line through the geometric axis of the reticle, shall be parallel to the image of a plumb line within ± 30 minutes of arc.

3.7.12 Exit pupil.- The minimum dimension of each exit pupil shall be 0.180 inch for the low power system and 0.090 inch for the high power system.

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3.7.13 Eyepiece focus.- Eyepiece focus shall be adjustable over the range of plus and minus 4 diopters as read on each eyepiece scale. The zero diopter setting of each eyepiece scale shall indicate zero diopter within ± 0.5 diopter when infinity target images are brought into best focus using the high and low power positions.

3.7.14 Interpupillary adjustment.- The eyepiece shall be adjustable between 58 and 72 millimeters. The interpupillary scale shall indicate the correct interpupillary distance at 65 millimeters within 0.5 millimeter.

3.7.15 Diopter focus.- The requirements of 3.7.15.1 to 3.7.15.3 inclusive shall be met in the low power position only.

3.7.15.1 Periscope reticle.- The difference in focus between the horizontal and vertical lines of the reticle image shall not exceed 0.25 diopter as read on periscope diopter scale.

3.7.15.2 Elevation scale.- The difference in focus between the center of the periscope reticle and the elevation scale shall not exceed 0.5 diopters when viewed with the unaided eye.

3.7.15.3 Azimuth scale.- The difference in focus between the center of the periscope reticle and the azimuth scale shall not exceed 0.5 diopter when viewed with the unaided eye.

3.7.16 Scale exit pupils.- The minimum dimensions of the exit pupil of the azimuth and elevation projection system shall be 0.070 inch.

3.7.17 Azimuth scale index.- The azimuth scale index shall be horizontally centered in the aperture opening within 2.0 mils.

3.7.18 Elevation scale index.- The elevation scale index shall be vertically centered in the aperture opening within 2.0 mils.

3.7.19 Scale markings.- Azimuth and elevation scale index markings shall overlap the edge of the mil scale from 1/3 to 2/3 the length of a small scale graduation.

3.7.20 Scale presentations.- The scale and scale index markings shall be parallel within one degree to the lines of the periscope reticle.

3.7.21 Parallax of scale graduations.- Parallax between scale graduations and the scale indexes shall not exceed the width of one scale graduation.

3.7.22 Elevation travel.- The line of sight shall travel through an excursion from at least 500 mils below a horizontal line of sight to at least 800 mils above a horizontal line of sight.

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3.7.23 Elevation accuracy.- The elevation scale readings shall indicate the true change in elevation of the line of sight within 0.3 mil.

3.7.24 Plumb travel.- The line of sight shall be in coincidence with a plumb line target within 0.3 mil over the total excursion of elevation travel starting at 500 mils depression. In addition the total spread between any two error readings shall not exceed 0.3 mil.

3.7.25 Azimuth displacement.- The azimuth shift in the line of sight at zero elevation shall not exceed 0.2 mil after an elevation change of 50 mils and return to zero elevation is made.

3.7.26 Elevation displacement.- The elevation shift in the line of sight at zero azimuth shall not exceed 0.2 mil after an azimuth change of 100 mils and return to zero azimuth is made.

3.7.27 Lift.- The line of sight shall not vary by more than 0.2 mil from an established elevation setting when the periscope is rotated 6400 mils in azimuth in one direction, without reversal.

3.7.28 Azimuth accuracy.- The azimuth scale readings shall indicate the true change in deflection of the line of sight within 0.3 mil.

3.7.29 Compass.

3.7.29.1 Needle lock.- Depression of the needle lock shall permit free and unrestricted movement of the compass needle within the confines of the housing. Release of the lock shall immobilize the needle off the pivot.

3.7.29.2 Magnetic attraction.- With the needle lock depressed the compass needle shall point north and shall be centered in deflection within the compass housing when the telescope level vials are centered.

3.7.29.3 Balance.- With the needle lock depressed the compass needle shall balance to bring its two extremities at equal clearance from the inside bottom of the compass housing when the telescope level vials are centered.

3.7.29.4 Sealing.- The interior of the compass shall show no evidence of dirt or moisture.

3.7.30 Leveling screw excursion.- Each leveling screw knob shall be capable of being rotated 9 turns ± 1/2 turn.

3.7.31 Operability.- The periscope shall meet the applicable operability requirements of Tables I and II when checked at the temperature indicated therein. Torque values of Table I are maximum for breaking torque, (maximum torque to start motion) and rotation, to maximum

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clockwise and counterclockwise limits of motion.

TABLE I

		Torque Unit of Measure	-40°F	+60 to 90°F	+150°F
1.	Instrument slewing	in. lbs.	40	20	20
2.	Azimuth knob	in. oz.	40	20	20
3.	Azimuth scale knob	in. oz.	40	20	20
4.	Elevation knob	in. oz.	35	30	30
5.	Leveling screw knobs (3) *(a)	in. lbs.	10	7	7
6.	Magnification selector	in. lbs.	20	15	15
7.	Filter selector	in. oz.	75	60	60
8.	Left eyepiece, angular movement *(b)	in. lbs.	10	5	5
9.	Diopter adjustment left & right	in. oz.	40	30	30
10.	Scale illuminator azimuth	in. oz.	60	50	50
11.	Scale illuminator elevation	in. oz.	60	50	50
12.	Mirror (rotation) azimuth	in. oz.	60	50	50
13.	Mirror (rotation) elevation	in. oz.	60	50	50
14.	Mirror (angular) azimuth	in. oz.	25	20	20
15.	Mirror (angular) elevation	in. oz.	25	20	20

*(a) The difference in torque readings between the 3 leveling screw knobs shall not exceed 2 inch pounds as read at temperature of between +60° and + 90°F only

*(b) With the interpupillary lock in released position

TABLE II

The following items shall function as operable at each temperature, -40°F, +60 to 90°F and +150°F.

	ITEM	FUNCTION
1.	Toggle switch	Off-On
2.	Compass needle lock	Release and lock compass needle
3.	Level vial covers	Rotate to expose vial Rotate to protect vial
4.	Azimuth scale knob cover	Open, close
5.	Electric input plug	From disconnected position connect with no intermittence of continuity

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TABLE II (Continued)

ITEM	FUNCTION
6. Eyepiece shields	Soft and pliable
7. Magnification selector lever	Lock in detent 10X and 20X position
8. Head sun shield cover	To open and close and remain in any set position over its total excursion

3.7.32 Electrical.- The electrical wiring of the periscope shall conform to the wiring diagram and any notes thereon.

3.7.33 Continuity.- Each conductor of the periscope shall be electrically continuous (nominally 0 resistance) between appropriate terminations as specified on the wiring diagram.

3.7.34 Insulation resistance.- Insulation resistance between all mutually insulated points and ground shall not be less than 1000 megohms when subjected to a potential of 500 volts direct current.

3.7.35 Illumination.- The application of 3 volts to the electrical input socket shall provide illumination for the telescope reticle, and the azimuth and elevation scales to be clearly visible when the two potentiometers are in minimum resistance (counterclockwise) position. The two tubular level vials shall be illuminated by actuation of the toggle switch independent of the two potentiometers.

3.7.35.1 Stray light.- There shall be no indication of stray light visible from any part of the periscope when viewed at a distance of 45 to 50 meters in a dark environment of 10^{-5} foot candles of illumination.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

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4.2 First article submission.- When a first article sample is required by the contract it shall be submitted to the Government for test and evaluation in accordance with the contract schedule. The sample shall consist of three (3) periscopes. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, equipment, processes and including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply. After provisional acceptance at source, the initial production samples will be inspected and tested by the Government for all requirements and tests of the drawings and specifications. Government inspection and testing may be conducted at the contract facility or as otherwise specified by the contract. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and sub-assemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables III, IV and V herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables III, IV and V shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE III - CLASSIFICATION OF DEFECTS

CLASS CRITICAL: NONE MAJOR: AQL 0.65% Defective	REQUIREMENT	TEST PROCEDURE
101. Vibration	3.5.3	4.6.3
102. Sealing & flushing	3.5.4	4.6.4
103. Cleanliness	3.6	4.6.5
104. Resolution	3.7.5	4.6.6
105. Level vial readings	3.7.1.1	4.6.7.1

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TABLE III - CLASSIFICATION OF DEFECTS (Continued)

CLASS CRITICAL: NONE MAJOR: AQL 0.65% Defective	REQUIREMENT	TEST PROCEDURE
106. Collimation	3.7.2	4.6.7.2
107. Level line	3.7.3	4.6.7.3
108. Field of view	3.7.4	4.6.7.4
109. Parallax	3.7.6	4.6.7.5
110. Magnification	3.7.7	4.6.7.6
111. Filters	3.7.8	4.6.7.7
112. Reticle accuracy	3.7.9	4.6.7.8
113. Image tilt	3.7.10	4.6.7.9
114. Parallelism of reticle & image	3.7.11	4.6.7.10
115. Exit pupils	3.7.12	4.6.7.11
116. Eyepiece focus	3.7.13	4.6.7.12
117. Interpupillary adjustment	3.7.14	4.6.7.13
118. Diopter focus, periscope reticle	3.7.15.1	4.6.7.14
119. Diopter focus, elevation scale	3.7.15.2	4.6.7.14.1
120. Diopter focus, azimuth scale	3.7.15.2	4.6.7.14.2
121. Scale exit pupils	3.7.16	4.6.7.15
122. Azimuth scale index	3.7.17	4.6.7.16
123. Elevation scale index	3.7.18	4.6.7.17
124. Scale markings	3.7.19	4.6.7.18
125. Scale presentations	3.7.20	4.6.7.19
126. Parallax of scale graduations	3.7.21	4.6.7.20
127. Elevation travel	3.7.22	4.6.7.21
128. Elevation accuracy	3.7.23	4.6.7.22
129. Plumb travel	3.7.24	4.6.7.23
130. Azimuth displacement	3.7.25	4.6.7.24
131. Elevation displacement	3.7.26	4.6.7.25
132. Lift	3.7.27	4.6.7.26
133. Azimuth accuracy	3.7.28	4.6.7.27
134. Compass, needle lock	3.7.29.1	4.6.8
135. Compass, needle attraction	3.7.29.2	4.6.9
136. Compass, needle balance	3.7.29.3	4.6.10
137. Compass, sealing	3.7.29.4	4.6.11
138. Leveling screw excursion	3.7.30	4.6.12
139. Operability	3.7.31	4.6.13
140. Electrical	3.7.32	4.6.14
141. Continuity	3.7.33	4.6.15
142. Insulation resistance	3.7.34	4.6.16
143. Illumination	3.7.35	4.6.17
144. Stray light	3.7.35.1	4.6.18
145. Workmanship	3.4	4.6.19
MINOR: NONE		

NOTE: The tests in Table III shall be conducted at a temperature between +60° and +90°F.

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4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One periscope shall be selected at random as a special sampling from each 10 produced. The sample shall meet the requirements and tests in Table IV.

TABLE IV

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable Drawing - Visual
302.	General specification	3.2	MIL-F-13926 - Visual

NOTE: The tests in Table IV shall be conducted at a temperature between +60° and 90°F.

4.4.2 Environmental.

4.4.2.1 Periscope.- Three periscopes shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall have met the requirements and tests in Table V and shall then meet the requirements and tests in Table III.

TABLE V

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Storage temperature (-65° and +165°F)	3.5.1	4.6.1
304.	Extreme operating temperatures (Torque, -40° and +150°F)	3.5.2, 3.7.31	4.6.2, 4.6.13
305.	Extreme operating temperatures (Functional operability, -40° and +150°F)	3.5.2, 3.7.31	4.6.2, 4.6.13

4.4.3 Failure of sample.- Should any item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

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4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contra all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Performance of periscope assembly.- Special testing equipment to test the periscope assembly shall conform to the following requirements (see Figure 1):

(a) Azimuth test table.- Provisions shall be made for centering the base of the periscope relative to the rotary table axis and locking the periscope base in this position on the rotary table. The azimuth test table shall indicate azimuth rotation of the periscope to tenths of a mil, with an accuracy of .03 mil, throughout full azimuth excursion. During azimuth rotation the mounting surface utilized for the periscope base shall remain level within 5 seconds (.025 mils).

(b) Target collimators.- All target projector collimators shall have a clear aperture of at least 1.0 inch and contain appropriate reticle patterns for measuring the mechanical and optical accuracies of the periscope. The target collimators shall be set at infinity focus, unless

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otherwise specified. All zero line of sight collimators shall be set with the lines of sight in the same horizontal plane (approximately 22.50 inches above the periscope mounting surface) within 5 seconds. The lines of sight of target collimators "A", shall be parallel to each other. The elevation and depression target collimators lines of sight shall be positioned in the same vertical plane with the zero line of sight target collimator "B". This vertical plane shall be perpendicular to the periscope mounting surface within 5 seconds. The angular positions of the lines of sight of the elevation and depression target collimators shall be positioned at depressions of -500 mils, -250 mils and elevations of +250 mils, +500 mils and +800 mils with respect to the periscope mounting surface to an accuracy of .03 of a mil.

(c) Viewing double collimator.- The emergent rays of the periscope shall be measured with the aid of a double collimator aligned to the geometric optical axis of the periscope eyepiece assembly. The viewing collimators shall contain appropriate reticles for measuring collimation of the periscope. The viewing collimator lines of sight shall be parallel and lay in the same horizontal plane. The vertical reticle lines of the collimators shall be aligned to plumb lines when in the test position.

4.5.2.3.2 Sealing periscope.- Special testing equipment to test sealing of the periscope assembly shall conform to the internal pressure test requirements of Drawing F8565556, Pressure Tester-Two Station.

4.6 Test methods and procedures.

4.6.1 Storage temperature.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and conditions of 3.5.1. The periscope assembly shall be exposed to each temperature extreme specified in 3.5.1 for a minimum of 6 hours. After the temperature exposures, the periscope assembly shall be stabilized to an ambient temperature between +60° and +90° Fahrenheit and subjected to the appropriate visual and tactile examination of 4.6.3.1. The periscope shall conform to the requirements of 3.5.1. Caution shall be exercised during this test to avoid subjecting the periscope to thermal shock.

4.6.2 Extreme operating temperatures.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and conditions of 3.5.2. The periscope assembly shall be exposed to each temperature extreme specified in 3.5.2 for a period of 6 hours. At the end of each 6 hour period, and while still at the temperature extreme, the periscope shall be subjected to the applicable test of 4.6.13, to determine conformance to the requirements of 3.7.31 at the temperatures indicated. Caution shall be exercised during this test to avoid subjecting the periscope to thermal shock.

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4.6.3 Vibration.- Standard vibrating equipment capable of providin the amplitude and frequency as specified in 3.5.3 shall be utilized for the vibration test. Unless otherwise specified the test equipment shall be in accordance with MIL-F-13926. The periscope assembly shall be vibrated in the direction and conditions specified in 3.5.3. Upon completion of the vibration test the periscope assembly shall be subjecte to the visual and tactile examination specified in 4.6.3.1.

4.6.3.1 Examination.- Subsequent to vibration, the periscope shall be thoroughly examined for physical defects prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts. A tactile examination shall also be conducted to determine looseness or improper functioning of moving parts such as diopter movement, elevation and azimuth knob and magnificat selection lever. An examination shall also be made for loose internal parts or broken optical components, such as (lenses, prisms, head mirror, scales and reticle). Anyone of the possible defects defined above shall be cause to consider a periscope defective. Periscope assemblies which have passed the above examinations shall be subjected to the tests specified in 4.6.4 through 4.6.19 inclusive.

4.6.4 Sealing and flushing.- Internal pressurizing of the periscop to test for sealing shall be performed using nitrogen gas having a dew point at least as low as -25°F, and special testing equipment conforming to 4.5.2.3.2. The periscope shall be pressurized to the requirements of 3.5.4 in accordance with the procedure for internal pressure testing o Drawing F8565556. Maintain the applied pressure for the period specified to balance pressure within the periscope cells. With the pressure at 5 psig the periscope shall retain this pressure for 5 minutes ± 15 second with no indication of leakage. The pressure shall then be reduced to zero psig (normal atmospheric pressure). The periscope shall then be flushed with nitrogen to the requirements of 3.5.4. The interior atmosphere of the periscope shall be periodically sampled through a dew point meter until the atmosphere shows a dew point at least as low as -25°F. When this atmosphere is attained within the periscope, the periscope shall be sealed with an internal pressure of nitrogen at a pressure of between 0.25 and 0.50 psig at room temperature.

4.6.5 Cleanliness.- The optical surfaces of the periscope assembly shall be inspected for cleanliness in accordance with MIL-O-13830. The periscope assembly shall conform to the cleanliness requirements of 3.6.

4.6.6 Resolution.- Position the periscope on a holding device capable of accepting the mounting surface of the periscope. The test shall be performed utilizing an auxiliary telescope with a magnification of at least 3 power (3X) and a resolving power chart on a wall target. The test methods and procedures shall be in compliance with the "Resolutic Test" specified in MIL-O-13830 and the resolving power chart shall contain four line sets as shown in Figure I therein for the 10 power position. The auxiliary telescope shall be positioned at each eyepiece assembly of

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the periscope. The focus of the auxiliary telescope and eyepiece shall be established at one best focus setting for viewing of the detectable line structure and proper line count of the resolution target. Resolution of each optical system shall meet the requirements specified in 3.7.5. When the periscope is subjected to the 20 power resolution test the resolving power chart image may be outside the range of eyepiece adjustmen In this case remove the auxiliary telescope and utilize a single lens magnifier (5X) positioned close to the eyepiece in the plane of the exit pupil to focus on the resolution power chart image. The eyepiece may alsc adjusted to obtain one best focus setting for viewing of the detectable li structure and proper line count of the resolution power chart image. Resolution of each optical system shall meet the requirements specified in 3.7.5.

4.6.7 Orientation.- Perform the following tests with the periscope assembly positioned on special testing equipment conforming to 4.5.2.3.1.

4.6.7.1 Level vial readings.- The cylindrical level vials shall be centered utilizing the leveling knob screws of the periscope. The level vials shall meet the requirements specified in 3.7.1 and 3.7.1.1 when the periscope is positioned on the azimuth test table of the special testing equipment.

4.6.7.2 Collimation.- Perform this test at 10 and 20 power positio with the aid of the viewing double collimator positioned at the eye- piece of the periscope. With the filters of the periscope in clear position, set the diopter scales of the periscope to indicate zero and the interpupillary distance knob (IPD) positioned at 65 millimeters (MM). View through the right eyepiece of the viewing double collimator. Adjust the collimator in elevation or depression to establish coincidence of the intersecting point of the zero vertical and horizontal reticle lines of the collimator with respect to the intersecting point of the zero vertica and horizontal target image lines. Observe through the left eyepiece of the collimator, the left zero vertical target imate line shall appear on the side of the zero vertical reticle line of the collimator which repres divergence of the target images. Measure the distance from the right target image to the left target image in elevation (dipvergence). The emerging rays from the two eyepieces shall be within the tolerance specif in 3.7.2. Record the divergence readings. Repeat the above procedure with the IPD knob positioned at 58MM and 72MM setting. Record the divergence and dipvergence readings. The requirements specified in 3.7.2 shall be met.

4.6.7.3 Level line.- Observe through the right eyepiece of the periscope. The line of sight through the geometric axis of the periscope reticle pattern shall meet the requirements specified in 3.7.3.

4.6.7.4 Field of view.- The centering error of the field of view (reticle pattern) in the right eyepiece relative to the right diaphragm shall be measured by noting the reticle graduation adjacent to the diaphr

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at the two extreme elevation and azimuth positions. The difference between the two readings shall not exceed twice the centering error specified in 3.7.4.

4.6.7.5 Parallax.- Perform this test with the right diopter knob scale of the periscope set for best focus of the periscope reticle. Parallax requirements shall be met when the right target image is adjusted as specified in 3.7.6. Parallax will be recognized by any apparent displacement of the periscope reticle center lines with respect to the center target image lines when the observer's head is moved from side to side or up and down. Displacement between the periscope reticle and target image shall not exceed the tolerance specified in 3.7.6.

4.6.7.6 Magnification.- The right and left optical systems of the periscope shall be inspected for magnification utilizing a dynameter and a diaphragm with a known diameter. Position the diaphragm at the entrance window of the periscope incident to the right objective. Set the dynameter at the right eyepiece. While viewing through the dynameter, adjust the dynameter for best focus of the diaphragm image. Measure the diameter of this image utilizing the millimeter reticle scale of the dynameter. Divide the known diameter by the diaphragm image diameter as read on the dynameter reticle scale. Record this value. Repeat the magnification test procedure above for the left optical system. Record this value. Any difference between the two recorded values shall not exceed the tolerance specified in 3.7.7 as measured at the low and high magnification positions.

4.6.7.7 Filters.- Perform this test in sequence with the test in 4.6.6.6. Position the filter knob to read (5) and view through the eyepiece assembly of the periscope. The clear filter shall be in the field of view with no overlap of adjacent filters. Introduction of the remaining filters into the optical system shall meet the requirements specified in 3.7.8.

4.6.7.8 Reticle accuracy.- This test shall be conducted in accordance with test for reticle scale spacings of MIL-O-13830. The total angular subtense of the reticle graduations shall meet the requirements specified in 3.7.9.

4.6.7.9 Image tilt.- Perform this test with the aid of an auxiliary telescope of at least 3 power (3X) having a vertical reticle line. A qualified leveled surface shall be utilized for positioning the auxiliary telescope at each eyepiece of the periscope. Plumb the vertical line of the auxiliary telescope. Position the auxiliary telescope at the right eyepiece of the periscope. Observe through the auxiliary telescope and establish one extremity of the auxiliary telescope vertical reticle line coincident to a vertical target image line. The vertical reticle line of the auxiliary telescope with respect to the vertical target image line shall be within 1.6 mils at the edge of the field when measured over a field of view of 80 mils. Repeat the above procedure with the auxiliary telescope positioned at the left eyepiece of the periscope. The images

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formed by the two optical systems of the periscope shall meet the requirements specified in 3.7.10.

4.6.7.10 Parallelism of reticle and image.- Using the unaided eye position one extremity of the periscope vertical reticle line coincident to a vertical target image line. The reticle line shall be parallel to the vertical target image line within 0.8 mils at the edge of the field when measured over a field of view of 80 mils.

4.6.7.11 Exit pupils.- A dynameter shall be utilized for measuring the minimum dimensions of each exit pupil of the periscope. Position the dynameter at each eyepiece of the periscope. The exit pupils for the low and high power magnification shall meet the requirements specified in 3.7.12 as measured on the dynameter reticle scale.

4.6.7.12 Eyepiece focus.- A calibrated dioptometer with a magnification of at least 3 power (3X) shall be utilized for measuring the accuracy of the right and left zero diopter settings of the diopter scales. Set the dioptometer objective scale to zero and adjust the dioptometer eyepiece for best focus of the dioptometer reticle. Set the right diopter knob of the periscope eyepiece assembly to indicate zero diopter. Positio the dioptometer at the eyepiece of the periscope. Observe through the dioptometer, obtain best focus of the right target image by adjustment of the periscope eyepiece diopter knob. Repeat the above test procedure for best focus of the left target image utilizing the left eyepiece assembly. Each target image shall be in best focus within the tolerance specified in 3.7.13, as measured on the diopter knob scales of the periscope. Rotat each diopter knob from the established best focus of the target images to its clockwise stop, note the diopter setting on the scales, then rotate each diopter knob to its counterclockwise stop, note diopter setting on the scales. Each diopter movement shall meet the requirements specified in 3.7.13.

4.6.7.13 Interpupillary adjustment.- The interpupillary distance between each eyepiece shall be measured by means of the exit pupil diameters (edge to edge) utilizing a millimeter scale. The interpupillary scale shall meet the requirements of 3.7.14.

4.6.7.14 Diopter focus, periscope reticle.- Perform this test while observing through the right eyepiece of the periscope with the unaided eye and obtain best focus of the periscope reticle by adjustment of the periscope eyepiece diopter knob. Any difference in focus between the horizontal and vertical lines of the reticle image shall not exceed the tolerance specified in 3.7.15.1, as measured on the diopter knob scale. Record this measurement.

4.6.7.14.1 Diopter focus, elevation scale.- With the unaided eye obtain best focus of the elevation scale utilizing the diopter knob. The difference in focus between the recorded periscope reticle measurement

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and elevation scale shall meet the requirements specified in 3.7.15.2, as measured in the diopter knob scale.

4.6.7.14.2 Diopter focus, azimuth scale.- Perform this test utilizing the test procedure specified in 4.6.7.14.1. The azimuth scale shall meet the requirements specified in 3.7.15.3, as measured on the diopter knob scale.

4.6.7.15 Scale exit pupils.- A dynameter shall be utilized for measuring the dimension of the exit pupils of the azimuth and elevation scales. The exit pupils shall meet the requirements specified in 3.7.16 as measured on the dynameter reticle scale.

4.6.7.16 Azimuth scale index.- The azimuth scale index shall be horizontally centered in the aperture opening. The number of mils visible within the aperture opening shall be equally divided on each side of the index within the tolerance specified in 3.7.17.

4.6.7.17 Elevation scale index.- The elevation scale index shall be vertically centered in the aperture opening. The number of mils visible within the aperture opening shall be equally divided on each side of the index with the tolerance specified in 3.7.18.

4.6.7.18 Scale markings.- The azimuth and elevation scale index markings shall meet the requirements specified in 3.7.19, while viewing through the periscope eyepiece assembly.

4.6.7.19 Scale presentations.- Parallelism of the scale and index markings with respect to the periscope reticle is a visual test. Scale presentations shall meet the requirements specified in 3.7.20.

4.6.7.20 Parallax of scale graduations.- Perform this test while observing through the right eyepiece of the periscope and obtain best focus of the azimuth and elevation scales. Parallax will be recognized by any apparent displacement of the scale graduations with respect to the scale indexes when the observer's head is moved from side to side or up and down. Parallax of the scale graduations shall meet the requirements specified in 3.7.21.

4.5.7.21 Elevation travel.- With the line of sight of the periscope depressed to minus 500 mils below the (horizontal) zero line of sight, observe through the right eyepiece of the periscope and establish coincidence through the geometric axis of the reticle with the minus 500 mil horizontal target image line. Elevate the line of sight through the geometric axis of the reticle to plus 800 mils above the zero line of sight. The line of sight shall meet the requirements of 3.7.22.

4.6.7.22 Elevation accuracy.- With the line of sight through the boresight cross of the periscope reticle in coincidence with the images of targets established at the following depression and elevation setting in mils -500, -250, 0 (horizontal line of sight), +250, +500 and +800, the errors in elevation change between any two target points as determined from the scale readings shall not exceed the value specified in 3.7.23.

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4.6.7.23 Plumb travel.- While sighting through the right eyepiece, depress the line of sight of the periscope -500 mils below the (horizontal) zero line of sight. Establish coincidence of the center point of the boresight cross of the periscope reticle with respect to the vertical target image line. Elevate the line of sight to the following established settings in mils, -250, "0" (horizontal line of sight), +250, +500 and +800 mils. The center point of the boresight cross shall be in coincidence with the vertical target image line at each known elevation within the tolerance specified in 3.7.24. The total deviation between any two error readings shall not exceed the tolerance specified in 3.7.24.

4.6.7.24 Azimuth displacement.- Observe through the right eyepiece of the periscope and position the azimuth and elevation scales to zero mils utilizing the azimuth setting and elevation knobs of the periscope. The center point of the boresight cross of the periscope reticle shall be in coincidence with respect to the intersecting point of the zero horizontal and vertical target image lines of the "0" (horizontal line of sight). Rotate the elevation knob to elevate the line of sight 50 mils. Any azimugh displacement, between the line of sight with respect to the center point of the boresight cross, due to reversing the travel of the elevation knob to reestablish zero elevation shall not exceed the tolerance specified in 3.7.25 as measured on the periscope reticle scale.

4.6.7.25 Elevation displacement.- Establish coincidence of the center point of the boresight cross with respect to the intersecting point of the zero elevation horizontal and vertical target image lines. Rotate the azimuth setting knob to change the azimuth line of sight 100 mils. Any elevation displacement, between the line of sight with respect to the center point of the boresight cross, due to reversing the travel of the azimuth setting knob to reestablish zero azimuth position shall not exceed the tolerance specified in 3.7.26 as measured on the periscope reticle.

4.6.7.26 Lift.- Lift accuracy readings shall be made at 400 mil increments. As each 400 mil position is established on the azimuth test table, rotate the periscope 400 mils by means of the periscope azimuth control knob to reestablish coincidence of the center point of the boresight cross with respect to the zero horizontal target image line at each 400 mil position throughout 6400 mils of azimuth rotation shall be recorded. The recorded lift displacement error readings shall not exceed the tolerance specified in 3.7.27 as measured on the elevation scale of the periscope.

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4.6.7.27 Azimuth accuracy.- The center point of the boresight cross shall be in coincidence with the zero vertical target image line, prior to performing the azimuth accuracy test. The azimuth readings shall be made at 400 mils increments. As each 400 mil position is established on the azimuth test table, rotate the periscope by means of the periscope azimuth knob to reestablish coincidence of the center point of the boresight cross with respect to the zero vertical target image line without overtravel. Record the azimuth readings as measured on the azimuth scale of the periscope at each 400 mil increment position throughout 6400 mils of azimuth rotation. Each recorded azimuth reading shall meet the requirements specified in 3.7.28.

4.6.8 Compass, needle lock.- Depression of the needle lock shall permit free and unrestricted movement of the compass needle. The compass needle shall meet the requirements specified in 3.7.29.1.

4.6.9 Compass, magnetic attraction.- Perform this test utilizing a calibrated compass and/or engineer's transit positioned on a non-magnetic leveled surface to establish a target point representative of magnetic north direction. Position the periscope on the surface plate and center the level vials as specified in 3.7.29.2, utilizing the periscope leveling screws. While observing through the right eyepiece, orient the periscope to obtain coincidence of the periscope vertical reticle line with respect to the established target point. The compass needle shall meet the requirements specified in 3.7.29.2, when the needle lock is depressed.

NOTE: The periscope shall be positioned on the leveled surface as close as possible to the original position of the calibrated compass and/or engineer's transit to assure the same magnetic influences are maintained.

4.6.10 Compass, needle balance.- Perform this test with the periscope positioned on a leveled surface and the level vials centered as specified in 3.7.29.3, utilizing the periscope leveling screws. With the needle lock depressed, observe through the compass viewer. The compass needle shall meet the requirements specified in 3.7.29.3.

4.6.11 Compass, sealing.- The compass shall be visually inspected to determine conformance to the requirements of 3.7.29.4.

4.6.12 Leveling screw excursion.- The leveling screw knobs shall be subjected to a visual and tactile examination. Rotation of each screw knob through its full excursion range shall meet the requirements specified in 3.7.30.

4.6.13 Operability.- Standard torque measuring equipment capable of measuring the torque values specified in 3.7.31, Table I shall be utilized for these tests. Appropriate adapters to accommodate the components under tests shall be used with the torque measuring equipment.

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Torque shall be measured while rotating the components through their full range of travel. The measured torque values shall meet the requirements specified in 3.7.31, Table I at the temperatures indicated. Subsequent to the torque tests and while still at the temperature extreme the periscope shall be subjected to the applicable functional operability tests to determine conformance to the requirements of 3.7.31, Table II at the temperatures indicated.

4.6.14 Electrical.- The wiring of the periscope shall be inspected for conformance to the wiring diagram and layout drawings utilizing standard measuring equipment to determine conformance with the requirements of 3.7.32.

4.6.15 Continuity.- A continuity test of the periscope shall be made utilizing standard measuring equipment to determine conformance to the electrical wiring drawing. The periscope shall meet the requirements specified in 3.7.33.

4.6.16 Insulation resistance.- The periscope shall be subjected to the insulation resistance test in accordance with Method 302, Test Condition B of MIL-STD-202. Measurement between all mutually insulated points shall be made utilizing standard measuring equipment to determine compliance with the requirements of 3.7.34.

4.6.17 Illumination.- Apply the operating voltage and conditions specified in 3.7.35 to the periscope assembly. The illuminated periscope reticle and azimuth and elevation scales shall be clearly defined when visually examined against a darkened background. Should the brightness of a reticle or scale defect be greater than the etched lines, the defect shall be cause for rejection. Actuation of the toggle switch shall provid illumination to the tubular level vials as specified in 3.7.35.

4.6.18 Stray light.- Perform this test with the periscope positioned in its normal operation position and the conditions specified in 3.7.35.1. The elevation and azimuth scales shall be illuminated for maximum brightness. With the sun shield in the open position, the elevation scale setti at -500 mils, the periscope shall meet the requirements specified in 3.7.35.1 throughout two complete revolutions of azimuth rotation of the periscope in a time duration of between two and three minutes. This test shall be performed only after the observers eyes have become dark adapted.

4.6.19 Workmanship.- The periscope assembly shall be subjected to a visual examination and shall meet the requirements specified in 3.4.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- The Periscope, Battery Command, M43 shall be placed in Case, Battery Command Periscope: M165, Part Number 10540496, Federal Stock Number 6650-885-5745. Packaging of the periscope and the case shall be in accordance with Packaging Data Sheet 8624251.

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The level of protection shall be as specified in the procurement document (see 6.2 and 6.3).

6. NOTES

6.1 Intended use.- The Periscope, Battery Command, M43 is a binocular periscopic theodolite type of instrument of optical mechanical assembly offering a selective magnification of 10 or 20 power. It is primarly intended to observe enemy targets and accurately measure azimuth and elevation angles from a known reference point or from magnetic north as established from the compass mounted on the periscope.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of applicable levels of preservation, packaging and packing
(c) Applicable stock number
(d) Applicable packaging data sheet (see 5.1)
(e) Availability and quantity required of Case, Battery Command Periscope: M165 (see 6.3)

6.3 Case, Battery Command Periscope: M165.- The contracting officer shall establish the availability of Case, Battery Command Periscope: M165 to the supplier of the Periscope B.C. M43. The case shall be fabricated to MIL-C-46355 and shall be utilized as the packing and shipping container for the periscope.

Custodian:

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Preparing activity:

Army - MU

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three mutually perpendicular axes.

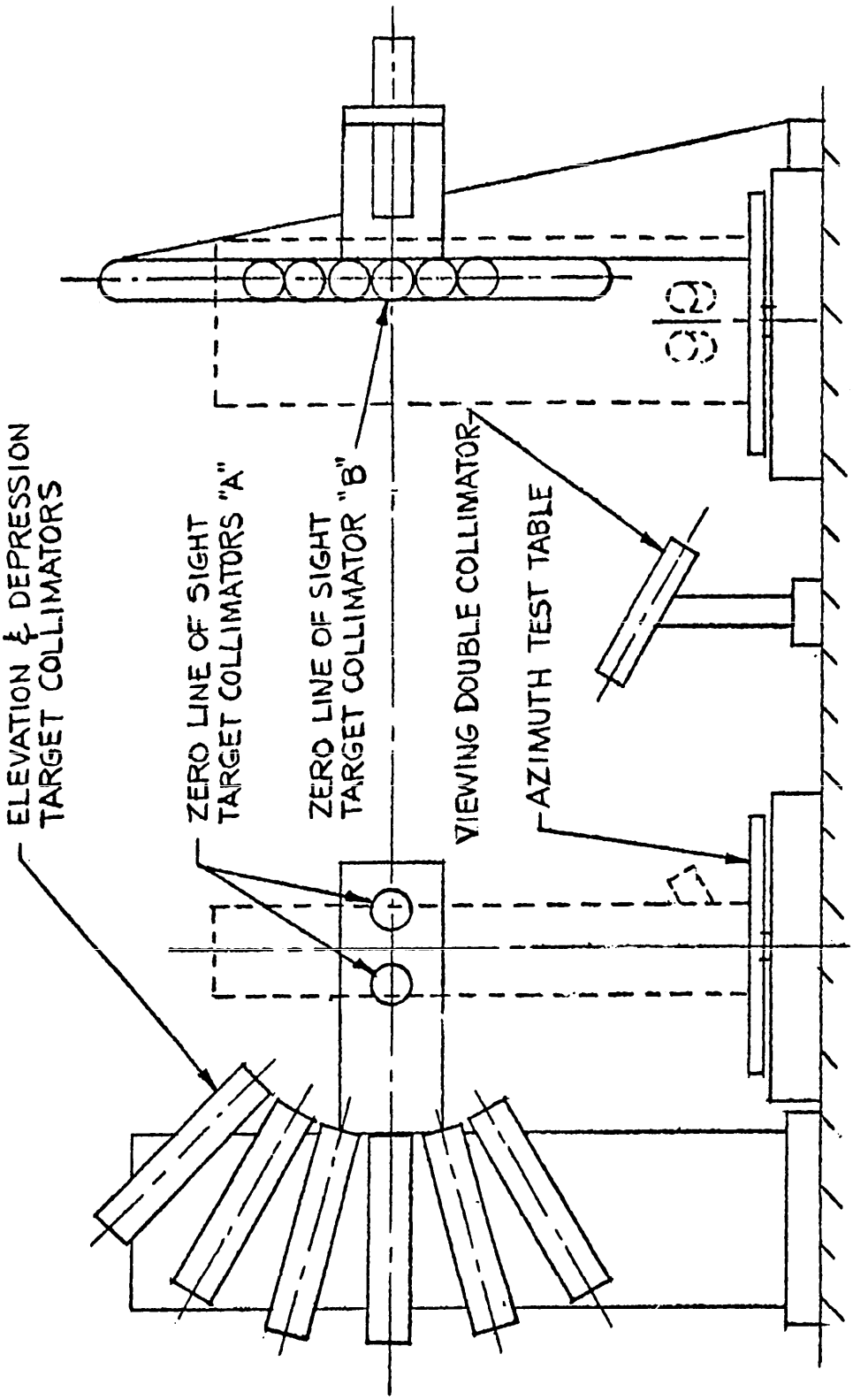


FIGURE 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R25

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in th use of the specification in procurement of products for ultimate use by the Department of Defense. This shee is provided for obtaining information on the use of this specification which will insure that suitable product can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Commen and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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